

Motor control: In constant pursuit of optimality

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Humans learn through exploration. A new study suggests that this may be how we learn to save energy when we walk.

Spend enough time watching a toddler roam the house during a global pandemic and one will eventually reach two conclusions: first, many of their behaviors appear hopelessly aimless (like trying to swim across the coffee table); and second, the absurd behaviors that engender the biggest laughs from their parents are repeated with outsized frequency. There is a subtle brilliance in the toddler's wild behavior: they leverage the apparent randomness of their silliness to discover new ways of maximizing positive reinforcement. If the nearby parent laughs and joins the playing — repeat this in the future! If the parent remains focused on the grant being written on their laptop — time to try something different (again, and again, and...). But this link between exploration and reinforcement is not limited to rambunctious toddlers: in a new study reported in this issue of *Current Biology*, Abram *et al.*¹ provide insight into how people leverage exploration to optimize their movements.

Reinforcement of movements that produce reward

A significant body of work on behavioral reinforcement dating back to seminal behaviorists like Pavlov, Thorndike, Watson, and Skinner helps to explain these patterns of behavior. As a result of this work, we know that behaviors that produce reward are more likely to be repeated in the future. We also know that there are many ways that these behaviors can be discovered and reinforced. Humans and other animals can learn associations between behavior and reward explicitly (for example, when a child helps to clean up toys and receives a snack afterward) or more implicitly through exploration (for example, when a

child learns to walk by falling repeatedly over time²).

From the perspective of exploration, there has been considerable interest in the role that movement variability plays in helping humans to discover better ways to move. Our movements are inherently variable — even the best athletes in the world cannot produce identical movements twice in a row — but it is generally accepted that this variability is not always ‘bad’, or purely unwanted noise. Instead, movement variability consists of both noise and a form of natural exploration, where the nervous system may subtly alter movement patterns in persistent search of a better way to move^{3–6}.

Many studies have suggested a link between one's natural movement variability and one's ability to learn new, potentially beneficial movement patterns in pursuit of optimization^{7–11}. However, we are still learning about *how* the nervous system regulates the interactions between variability and optimal control. This is an important question because our ability to understand how humans and animals leverage variability and exploration to optimize their movements and interactions with their surroundings has, in our view, considerable potential implications for our abilities to deliver effective motor rehabilitation and design usable prosthetic and robotic devices.

Leveraging motor variability to discover optimal movement patterns

In their new study, Abram *et al.*¹ suggest that humans may leverage natural variability in their gait to optimize their walking patterns. The authors used robotic ankle exoskeletons — powered robots that fit around the lower leg and

assist ankle movement — to perturb the natural walking patterns of healthy young adults. Then, they simply observed how the participants responded as they walked. Across several days of practice with the exoskeletons, they found that these participants adapted many aspects of their gait in response to the robotic assistance. These included both readily observable features of their gait (for example, step cadence) and others that required closer inspection (for example, lower leg muscle activity). They also observed that the participants initially exhibited high levels of variability in these gait parameters upon initial activation of the exoskeletons that then decreased over time as participants gained experience walking with the assistance — a behavioral phenomenon commonly observed during motor learning^{11–14}. Critically, the investigators provided intentionally minimal instructions to the participants about how to walk while wearing the exoskeletons; instead, the participants had to rely on intrinsic feedback to guide any changes in movement over time.

From here, Abram *et al.*¹ made two key findings. First, they observed that the different features of gait did not all adapt and decrease in variability at the same rate: the aspects of gait that adapted quickly also showed faster decreases in variability. Second, the adaptations and decreases in gait variability were largely associated with decreases in the metabolic cost of walking (the energy expended to walk). In other words, the gait parameters that changed quickly were associated with adoption of a more economical gait. Collectively, these findings suggest that the human nervous

system may leverage the natural variability in one's gait to explore for more economical solutions, and that this exploration may be biased in ways that help to reduce dimensionality in pursuit of this optimization.

Toward a comprehensive understanding of motor adaptation and optimization

The findings reported by Abram *et al.*¹ also highlight important open questions about *how* and *why* the nervous system adapts gait. With regard to the *how*, it will be key to understand the mechanisms that contribute to initial scaling up and subsequent decrease in variability. External perturbations (such as those delivered by devices like exoskeletons) cause an immediate change in movement and pose a novel control problem that likely engages more than one learning mechanism. For example, adaptation as described here may also involve sensory prediction error-based learning of an internal model — the nervous system's ability to make predictions about the sensory outcome of motor commands in the face of new predictable dynamics^{15,16}. We would expect that there is a need to be able to predict the sensory consequences of a given motor command to exert precision control of walking under novel mechanical conditions¹⁶. It will be interesting to know how this (and other learning mechanisms) may interact or contribute to the processes described in this work. It will also be important to understand what triggers the nervous system to begin the adaptation process described in this study. There are situations where variability alone is not always sufficient for initiating gait adaptation^{17,18}. Further investigation into understanding how the nervous system controls and selectively tunes variability could provide significant insights into the mechanisms of motor learning that drive these changes.

With regard to *why* the nervous system adapts gait, the authors highlight the importance of continued study into how multiple objectives may be at play in the cost function(s) that the nervous system aims to optimize. The Abram *et al.*¹ study highlights minimization of energy expenditure as an important cost function — a notion that is well-supported by decades of prior literature. There is a

need for an integrated view of movement optimization that also considers other costs — including those associated with stability or pain, likely in combination with energy expenditure — that could also influence how the nervous system selects a movement pattern. This will likely require an improved understanding as to how the nervous system senses and measures these possible costs, a goal toward which this author team has already made significant progress^{19,20}.

Effective motor rehabilitation not only requires an ability to identify what is wrong with a patient's movement, but also an understanding about how and why the patient moves in a particular way. There is little question that the future of rehabilitation will be informed by studies like this one by Abram *et al.*¹ that leverage clever experimental designs to uncover keen behavioral insights into human motor control. As we continue to learn about how humans adapt our movements in pursuit of optimization, one can only hope — for their parents' sake — that the toddler will at least occasionally adapt their behavior to optimize for something other than maximizing the attention of their parents.

DECLARATION OF INTERESTS

The authors declare no competing interests.

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